

Linux Course Track

From first shell prompt to production Linux administration.

Live cohorts · Real lab machines · RHCSA-aligned · Nixpert certificate



```
$ nixpert enroll --track linux
✓ Shell Scripting ..... ready
✓ Linux Troubleshooting ..... ready
✓ Complete Linux ..... ready
$ _
```

Track Overview

Three courses that take you from your first shell prompt to administering, automating, and debugging production Linux. Take them individually or as the Linux Career Bundle.

	Linux Shell Scripting	Linux Troubleshooting	Complete Linux
Duration	4 weeks	5 weeks	10 weeks
Live sessions	8	10	20
Level	Beginner to Intermediate	Intermediate	Beginner to Intermediate
Prerequisites	Basic command line comfort	Working Linux knowledge	None
Cert alignment	—	RHCSA topics	RHCSA
Capstone	Production-grade automation suite	Timed broken-lab gauntlet	Build and harden a production server
Launch price	₹ 2,999	₹ 3,999	₹ 10,999
List price	₹ 3,499	₹ 4,999	₹ 12,999

Every cohort includes

- Weekly live classes with Q&A, recorded and available for 12 months.
- Hands-on labs for every module, with 1:1 support when a lab fights back.
- Private community chat and weekly office hours.
- A graded portfolio capstone and a Nixpert completion certificate.
- Upgrade credit: alumni move to a bigger course or bundle by paying only the difference plus ₹ 500.

Complete Linux

10 weeks · 20 live sessions · Beginner to Intermediate · ₹ 12,999 (launch ₹ 10,999)

The flagship track: from first login to administering real servers. RHCSA-aligned, so your study time counts twice. Every week pairs concepts with lab machines you actually break and fix.

Audience	Format	Outcome
Freshers, support engineers, Windows admins, career switchers	2 live sessions / week + lab machines	Administer production Linux and sit the RHCSA confidently

Week 1 · Getting Started

- Distributions and where they run: RHEL, Ubuntu, and friends.
- The shell: navigation, files and directories, globbing, man pages, and getting help.
- Editors that matter: vim survival skills and nano.

Lab: Install a VM, navigate the filesystem, and complete a 30-task shell scavenger hunt.

Week 2 · Users, Groups, and Permissions

- Users, groups, /etc/passwd and shadow, password policy.
- Permissions, umask, special bits, and ACLs.
- sudo done right: granting just enough privilege.

Lab: Build a multi-user system with shared project directories and correct ACLs.

Week 3 · Processes and systemd

- Processes, signals, priorities, and job control.
- systemd: units, services, targets, and dependencies.
- Writing your own service unit and reading unit failures.

Lab: Package a small app as a systemd service with restart policy and resource limits.

Week 4 · Software Management

- Package managers: dnf and apt, repositories, and GPG keys.
- Building from source, alternatives, and dependency debugging.

Lab: Stand up a local repository mirror and install from it.

Week 5 · Storage: Disks, LVM, and RAID

- Partitioning, filesystems (ext4, XFS), mounting, and fstab.
- LVM end to end: PVs, VGs, LVs, resizing, and snapshots.
- Software RAID concepts and mdadm basics.

Lab: Grow a live filesystem with LVM, then recover from a deliberately failed disk.

Week 6 · Networking

- IP addressing, routing, and NetworkManager.
- DNS, /etc/hosts, and name resolution debugging.
- SSH: keys, config, tunnels, and hardening. Firewalls with firewalld and nftables.

Lab: Configure a two-VM network with key-only SSH and a locked-down firewall.

Week 7 · Logs, Scheduling, and Backups

- journald and rsyslog: finding the truth in logs.
- cron, at, and systemd timers.
- Backup strategies with tar, rsync, and snapshots.

Lab: Build a scheduled, logged, verified backup job with rotation.

Week 8 · Kernel and Performance

- Kernel parameters, sysctl, and tuned profiles.
- CPU, memory, and I/O fundamentals: top, vmstat, iostat, and sar.
- Swap, OOM killer, and resource limits.

Lab: Diagnose a slow lab server using a structured performance checklist.

Week 9 · Security and Hardening

- SELinux and AppArmor: modes, contexts, booleans, and real troubleshooting.
- Hardening checklist: services, ports, updates, and auditing.

Lab: Fix a web app broken by SELinux without disabling SELinux.

Week 10 · Containers and Capstone

- Containers on production Linux: podman and docker, images, volumes, and networking.
- Running containers under systemd.

Capstone

Build, secure, automate, and document a production-style server: storage layout, services, firewall, monitoring, and backups. Defended live in the final session.

Linux Shell Scripting

4 weeks · 8 live sessions · Beginner to Intermediate · ₹ 3,499 (launch ₹ 2,999)

Bash from fundamentals to production automation. You leave writing scripts that handle errors, log properly, and survive contact with real systems.

Audience	Format	Outcome
Admins, support engineers, QA, and DevOps beginners	2 live sessions / week + scripting drills	Ship reliable automation scripts for real servers

Week 1 · Bash Fundamentals

- Variables, quoting rules, command substitution, and exit codes.
- Conditionals with test and `[[ ]]`, arithmetic, and string operations.

Lab: Write a system info script that degrades gracefully across distros.

Week 2 · Control Flow and Functions

- Loops (for, while, until), case statements, and reading input.
- Functions, arguments, getopts, and return values.
- Arrays and associative arrays.

Lab: Build a menu-driven user management tool with argument parsing.

Week 3 · Text Processing and Robustness

- Pipelines mastery: grep, sed, and awk for real log analysis.
- Error handling: set -euo pipefail, traps, and signals.
- Temporary files, locking, and concurrency pitfalls.

Lab: Parse a 100 MB log file into a daily report, with tests for malformed input.

Week 4 · Scripts in Production

- Running scripts from systemd units and cron with correct environments.
- Logging, notifications, and idempotency.
- POSIX portability and shellcheck. When to graduate to Python or Ansible.

Capstone

A production-grade automation suite: health checks, backup, and report generation, scheduled via systemd timers, with logging and failure alerts.

Linux Troubleshooting

5 weeks · 10 live sessions · Intermediate · ₹ 4,999 (launch ₹ 3,999)

Structured debugging of real failures: boot, networking, services, and performance. Every session includes a deliberately broken lab machine. This is the course that builds on-call confidence.

Audience	Prerequisite	Outcome
Admins, SREs, support and NOC engineers	Working Linux knowledge	A repeatable method for diagnosing any Linux failure

Week 1 · Method Before Madness

- A structured troubleshooting method: observe, hypothesise, test, fix, verify.
- Evidence gathering: journalctl, dmesg, /proc, and /sys.
- The USE method for resource problems.

Lab: Triage three mystery failures using only logs, no hints.

Week 2 · Boot Failures

- The boot chain: firmware, GRUB, kernel, initramfs, and systemd targets.
- Rescue and emergency modes, chroot recovery, and fstab disasters.
- Root password resets and corrupted bootloader recovery.

Lab: Recover four unbootable VMs: bad fstab, broken GRUB, missing initramfs, full root disk.

Week 3 · Networking Failures

- Layered diagnosis: link, IP, routing, DNS, and firewall.
- Tools that tell the truth: ip, ss, ping, traceroute, dig, and tcpdump.
- Reading packet captures without fear.

Lab: Find and fix five staged network faults, from wrong subnet mask to DNS hijack.

Week 4 · Services and Performance

- Why services fail: units, dependencies, permissions, SELinux, and ports.
- Performance debugging: CPU, memory, disk I/O, with top, vmstat, iostat, sar, and strace.
- Disk full, inode exhaustion, and runaway processes.

Lab: Restore a crawling web server to normal latency and prove it with before/after data.

Week 5 · The Gauntlet

- Bringing it together: incident notes, runbooks, and clean handovers.
- Capstone preparation and review.

Capstone

A timed broken-lab gauntlet: six faulty machines, ninety minutes, and a written incident report for each fix. Graded on method as much as outcome.

Pricing and Enrollment

Transparent pricing, launch discounts for the first cohort, and an upgrade path that rewards staying with Nixpert.

Course	Duration	List price	Launch price (first cohort)
Linux Shell Scripting	4 weeks	₹ 3,499	₹ 2,999
Linux Troubleshooting	5 weeks	₹ 4,999	₹ 3,999
Complete Linux	10 weeks	₹ 12,999	₹ 10,999
Linux Career Bundle (all three)	Flexible	₹ 17,999	₹ 14,999

Offers and policies

- **Upgrade credit:** alumni of any course can move up by paying only the price difference plus ₹ 500.
- **Cert alignment:** Complete Linux maps to RHCSA objectives, so exam preparation is built in.
- **Corporate and team training:** custom batches for teams of 5 or more. Write to hello@nixpert.in for a quote.
- **Payment:** UPI, cards, and net banking. EMI available on the flagship course and the bundle.
- **Refunds:** full refund up to 48 hours after the first live session, no questions asked.
- **Access:** recordings, labs, and community access for 12 months from enrollment.

How to enroll

- Visit nixpert.in and submit the enrollment form with your chosen course.
- We reply with the next cohort dates and an invite to a free intro call.
- Reserve your seat. Cohorts are capped to keep labs and 1:1 support meaningful.

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